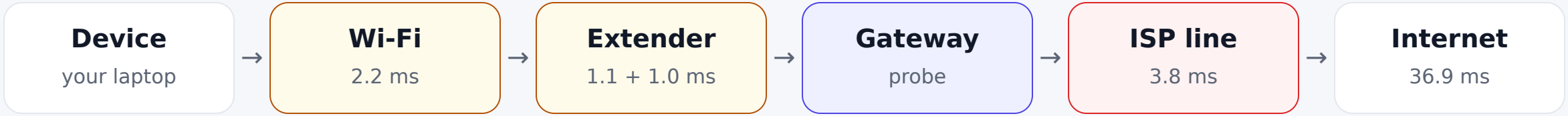
**Heimdall** latency health check for ISPs and their customers

# When the internet feels slow, who should fix it?

A small probe on the home gateway and the Wi-Fi extender measures every part of the path, and tells within seconds where the delay is and who should act.




**Probes on the gateway and the extender** · typical latency of each part on 30 Oct, from the daily analysis

## 1 · Measure

Every client connection's Wi-Fi and internet round trips, the extender link, the ISP's hops, 37 services, and the same services from AWS for comparison.

## 2 · Detect

Baselines per device and service. A slowdown becomes an incident with a likely cause: weak Wi-Fi, the extender, the ISP's line, its route to a service, DNS, or the service itself.

## 3 · Tell

A Slack alert, and a status page for the ISP and for the customer: what happened, why, who should act and the fix. Grafana for us, a weekly report for the home.

**No hostnames or server addresses are stored** · 8 MB static binary on a Zyxel EX7501 gateway and a WX5600 extender · AWS: Lambda, Firehose, Athena, DynamoDB, Grafana

**Heimdall** live demo

# Slow down my laptop, and watch it get caught

## 1 Slow it down `demo slow 40`

Adds 40 ms to my laptop's traffic. It is on the extender's Wi-Fi.

## 2 Caught in ~10 s `Slack`

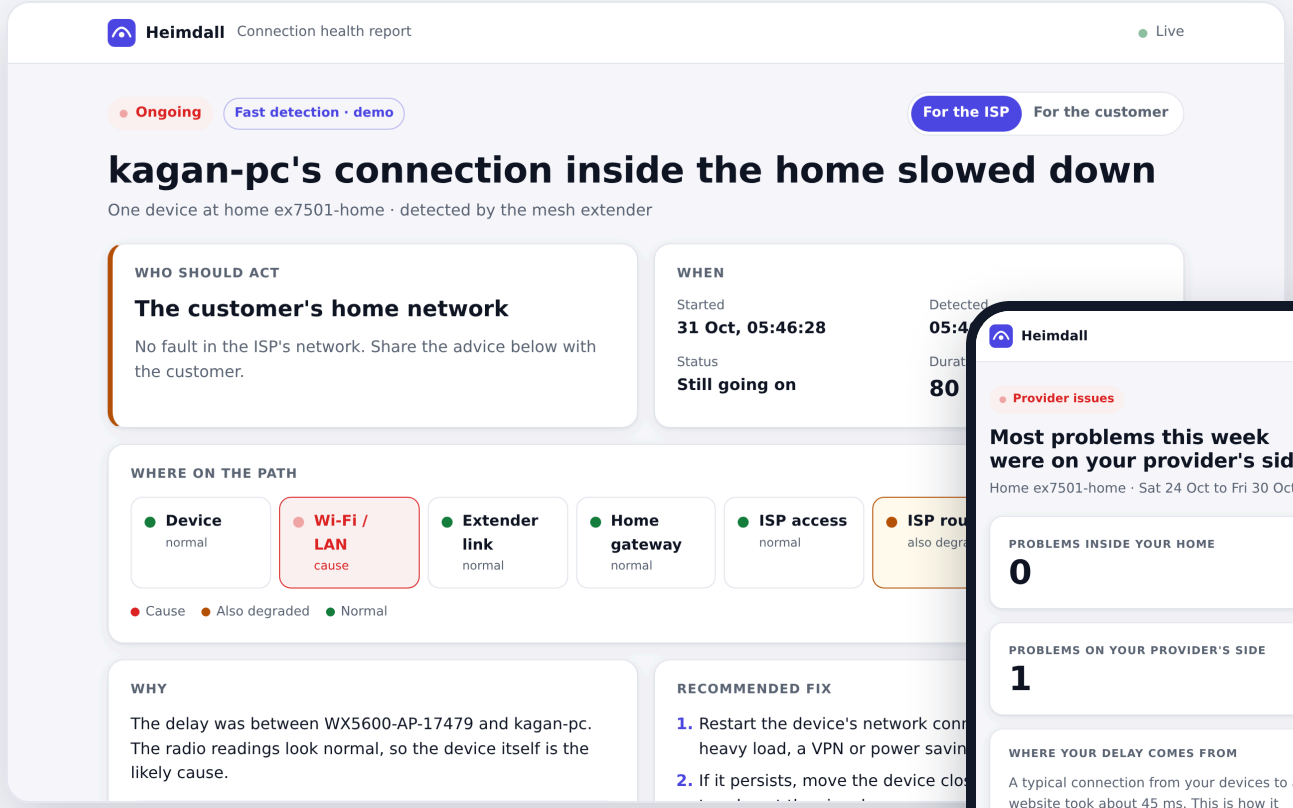
The extender's probe sees kagan-pc's hop jump from 1 ms to 41 ms.

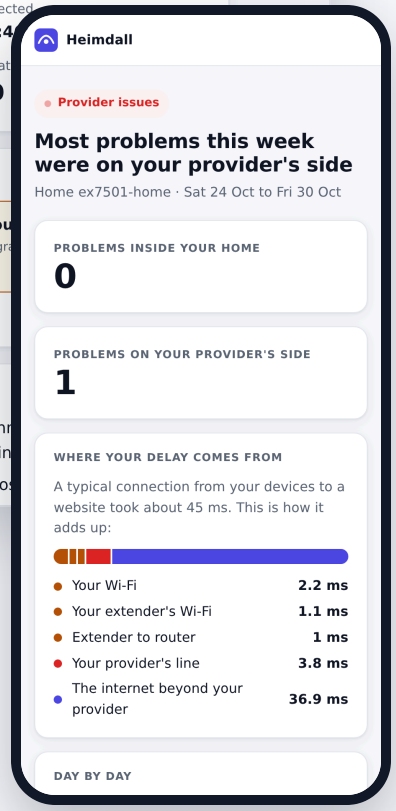
## 3 Who should act `status page`

The customer's home network, not the ISP: the ISP's line, its route and the services were normal.

## 4 Fix it `demo fix`

Resolved about 20 s later, with the real duration.




**For the ISP:** the cause, who should act, the fix
**For the customer:** the same in their words, and a weekly report of their home

Demo mode runs a second detector on 5-second buckets. The regular one waits 10 minutes before it alerts.

**Heimdall** can we trust it?

# We don't just measure, we cross-check

The gateway's numbers, checked against what a device measures and against the same services seen from the cloud.

**Device** heimdall-client on a laptop → **Gateway** the probe: where on the path → **Cloud** the same probe in AWS Frankfurt

## Device ↔ gateway

Does the gateway see what the device experiences?

**within ~3 ms** on 42,831 matched connections

| Service | Laptop | Gateway: internet + home | Gap |
|---|---|---|---|
| Google | 22.6 ms | 15.4 + 3.5 ms | **3.3 ms** |
| GitHub | 51.1 ms | 43.9 + 3.3 ms | **3.2 ms** |
| AWS Frankfurt | 55.2 ms | 48.4 + 3.3 ms | **3.2 ms** |
| YouTube | 27.1 ms | 17.5 + 3.6 ms | **3.4 ms** |

The steady gap is the laptop itself. Nothing to install on customers' devices.

## Gateway ↔ cloud

Is a slow service the ISP's fault or the service's?

**1,296** degraded service-hours over 6 days




slower only from the home: the ISP's line or route

slower from AWS too: the service

Most ISP-side: Microsoft 365, TRT Tabii, Gain, Exxen, Disney+. Each incident's status page shows the same comparison.

Matched by the client's address and port within 5 s · 27, 28, 31 Oct · each side is judged against its own normal